

BUSINESS REQUIREMENTS DOCUMENT

Tour & Activity Booking Platform with WhatsApp Commerce

AmwalPay Payment Gateway | WhatsApp Cloud API | Admin Control Panel

Prepared for: Project Owner

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1. Document Control

Field	Detail
Document Title	BRD – Tour & Activity Booking Platform with WhatsApp Commerce
Version	1.0 – Draft
Status	For stakeholder review; requirements to be validated before development sign-off
Prepared By	Claude (Anthropic), based on stakeholder idea/discovery notes
Distribution	Product Owner, Development Team / Agency, Payment Gateway Provider (AmwalPay), WhatsApp BSP

2. Executive Summary

This document defines the business and functional requirements for a production-grade, web-based Tour & Activity Booking Platform. The platform allows customers to browse tours/services, check real-time slot availability, book online, and pay either through the AmwalPay payment gateway or via a manual bank-transfer-with-screenshot workflow that is verified by an admin. It integrates deeply with the WhatsApp Cloud API so that customers can check availability, book, and track orders directly inside WhatsApp chat, supported by rich message templates (including media and carousels), a bot/automation builder, and an AI chat assistant. A full-featured admin panel gives the business owner complete control over bookings, orders, payments, staff, WhatsApp/email marketing, and reporting.

The system is designed around four pillars:

- **Booking Engine:** slot-based availability, multi-service booking, and order lifecycle management
- **Payments:** AmwalPay online gateway plus a manual bank-transfer approval workflow with screenshot proof
- **WhatsApp Commerce:** two-way conversational commerce — availability, booking, order status, notifications, broadcast marketing, and AI-assisted automation
- **Admin Control Center:** a single dashboard for operations, staff, CRM, marketing (email + WhatsApp) and analytics

3. Business Objectives

- Enable customers to discover, check availability, and book tours/services online in under 3 minutes.
- Reduce manual coordination by allowing bookings and order tracking directly through WhatsApp.
- Support both digital payments (AmwalPay) and manual bank transfer to accommodate customers without cards or with regional banking preferences.
- Give staff a shared inbox to manage all WhatsApp conversations without conflicts (assignment, labels, notes).
- Automate repetitive customer questions using an AI assistant and rule-based bot builder, escalating to a human when needed.
- Provide the business owner with a single admin panel with full visibility and control: bookings, revenue, staff performance, campaigns, and subscriber growth.
- Build a scalable base architecture that can add new tour categories, cities, currencies, and languages without a rebuild.

4. Project Scope

4.1 In Scope

- Public-facing responsive booking website (desktop + mobile web).
- Slot/availability engine and booking/order management system.
- AmwalPay online payment integration.
- Manual bank transfer payment method with screenshot upload and admin verification.
- WhatsApp Cloud API integration for availability check, booking, order status, notifications, broadcast, and templates (text, media, carousel).
- Bot/automation builder with AI assistant for WhatsApp conversations.
- Staff/team inbox: chat assignment, internal notes, labels/tags.
- Email system: template builder, contact sync, broadcast campaigns, subscriber management, transactional emails.
- Order confirmation notifications via Email + WhatsApp with full order detail.
- Full-featured admin/back-office panel with role-based access control.
- Reporting & analytics dashboard.

4.2 Out of Scope (Phase 1)

- Native mobile apps (iOS/Android) — website will be mobile-responsive/PWA-ready; native apps can be a future phase.
- Multi-vendor marketplace (multiple independent tour operators selling on one platform) — assumed single-business operation unless specified otherwise.
- In-house SMS gateway (WhatsApp is the primary conversational channel; SMS can be added later as fallback).
- Physical POS / on-site card terminal integration.

5. Stakeholders & User Roles

Role	Description
Customer / End User	Browses tours, checks availability, books, pays, and tracks orders via website or WhatsApp.
Super Admin	Full system control: settings, payment configuration, staff, integrations, all data.
Operations / Booking Admin	Manages tours, slots, pricing, bookings, order approvals, refunds.
Finance / Payment Verifier	Reviews manual bank-transfer submissions, approves/rejects with screenshot proof, reconciles payments.
Support / Chat Agent (Staff)	Handles assigned WhatsApp/email conversations, uses labels, escalates issues.
Marketing Manager	Builds email/WhatsApp templates, manages subscriber lists, runs broadcast campaigns, reviews performance.
Tour Guide / Field Staff (optional)	Views assigned bookings/schedule; can mark attendance/completion.

6. Functional Requirements

6.1 Public Booking Website

- Homepage with featured tours/services, search, and category/city filters.
- Tour/service detail page: gallery, description, itinerary, inclusions/exclusions, pricing tiers (adult/child/group), reviews/ratings.
- Real-time slot availability calendar per tour/service (date + time-slot level), showing remaining capacity.
- Multi-step booking flow: select date/slot → select quantity/pax → add-ons → customer details → payment method → confirmation.
- Guest checkout and registered-account checkout (login/OTP or email).
- Cart/multi-item booking (book more than one tour/date in a single order) — optional but recommended.
- Coupon/discount code and promotional pricing support.
- Multi-language support (e.g., English/Arabic) and multi-currency display (e.g., OMR base currency).
- SEO-friendly URLs, meta tags, sitemap, and page-speed optimized image delivery.
- Customer account area: booking history, invoices/receipts, order status tracking, rebooking.

6.2 Slot & Availability Management

- Admin can define per-tour capacity: total slots per day, per time-slot, or per guide/vehicle resource.
- Auto-block/close-out when a slot reaches full capacity, with optional waitlist.
- Blackout dates, seasonal pricing, and recurring schedule templates (e.g., daily 9 AM/2 PM departures).
- Overbooking protection: seat count is locked/held for a short window during checkout to prevent double-booking, released automatically if payment is not completed.
- Bulk slot creation/editing (e.g., generate 3 months of slots in one action) and bulk block-out.
- Live sync between website bookings and WhatsApp bookings — a seat booked from either channel instantly reduces available inventory on both.

6.3 Booking & Order Management

- Central Order Management System: every booking (from website or WhatsApp) generates a unique Order ID with full status lifecycle — Pending Payment → Payment Submitted → Verified/Paid → Confirmed → Completed / Cancelled / Refunded / No-show.
- Order detail view: customer info, tour/slot, pax count, add-ons, price breakdown, payment method, payment proof, staff notes, communication history (WhatsApp + email thread).
- Admin actions: approve/reject payment, modify booking (reschedule/date change), cancel with reason, issue full/partial refund, resend confirmation, generate invoice/voucher PDF.
- Order Management API — REST/GraphQL endpoints so other systems (accounting software, tour-guide app, reporting tools, future mobile app) can read/write order data securely (API key / OAuth2).
- Automatic reminders: pre-tour reminder (e.g., 24 hours before) via WhatsApp/email, and post-tour review request.
- Booking source tracking (Website / WhatsApp / Admin-manual / API) for reporting.

6.4 Payment Management

6.4.1 AmwalPay Payment Gateway (Online)

- Integrate AmwalPay hosted checkout / API for card and local payment methods (as offered by AmwalPay in Oman).

- Support authorization, capture, refund, and webhook/callback handling for payment status updates.
- Automatic order status update to "Confirmed" upon successful AmwalPay payment webhook, with instant email + WhatsApp confirmation.
- Handle failed/abandoned payments gracefully — retry payment link sent via email/WhatsApp.
- Payment reconciliation report matching AmwalPay settlement reports against internal orders.

6.4.2 Manual Bank Transfer + Screenshot Verification (Priority / Phase 1 Payment Method)

This is the payment flow explicitly requested as the first method to launch with:

- **Bank account setup:** Admin panel lets the business add one or more company bank accounts (bank name, account name, account number/IBAN, branch) that are displayed to the customer at checkout
- **Customer payment submission:** Customer selects "Bank Transfer", views the account details, makes the transfer, then submits a form with transaction reference/date and uploads a screenshot/photo of the payment proof
- **Pending state:** Order is created in "Payment Submitted – Pending Verification" status; customer and admin both notified instantly (admin gets a WhatsApp/email/dashboard alert)
- **Admin verification:** Verifier reviews the screenshot and reference in the order detail screen and clicks Approve or Reject (with a reason/comment for rejection)
- **Post-approval automation:** On approval, the order status changes to "Confirmed" and the system automatically sends a detailed confirmation via both Email and WhatsApp (order ID, tour, date/time, pax, amount paid, voucher/QR code)
- **On rejection:** Customer is notified with the reason and instructions to resubmit or contact support
- **Audit trail:** Every verification action is logged with the staff name, timestamp, and decision for accountability

6.4.3 Payment Method Framework (Future-Proofing)

- Design the payment layer as a pluggable module so additional gateways (e.g., Thawani, PayTabs, Apple Pay, STC Pay, or a second local PSP) can be added later without re-architecting.
- Support partial payment/deposit bookings (e.g., pay 25% now, balance before tour date) — suggested enhancement.

6.5 WhatsApp Cloud API — Conversational Commerce

This is the core differentiator of the platform. All the following are handled via the official WhatsApp Business Cloud API (Meta) through a Business Solution Provider (BSP) or direct Cloud API access:

6.5.1 Customer-Facing Conversational Features

- **Check availability in chat:** Customer types/selects a tour and date; bot replies with live slot availability pulled from the same engine as the website
- **Book directly in WhatsApp:** Guided conversational flow (interactive list/buttons) to select tour → date/slot → pax → add-ons → payment method → confirmation, without leaving WhatsApp
- **Pay in chat:** Send AmwalPay payment link, or accept bank-transfer screenshot directly as a chat attachment which auto-attaches to the order
- **Check order status/details in chat:** Customer sends order ID or "my orders"; bot replies with full detail (status, date, amount, voucher)
- **Rich media & carousel messages:** Product carousels (multiple tours with images, price, and a "Book Now" quick-reply button), image/video/PDF itinerary sharing, location pins
- **Order confirmation:** Full detailed confirmation message automatically sent after approval (booking summary, QR/voucher, support contact, cancellation policy)

6.5.2 Template Management

- Create, edit, and submit WhatsApp message templates (marketing, utility, authentication categories) for Meta approval directly from the admin panel.
- Support template variables/placeholders (customer name, order ID, date, amount, etc.), header media (image/video/document), and call-to-action / quick-reply buttons.
- Template library with status tracking (Draft / Pending Approval / Approved / Rejected) and version history.
- Sync approved templates automatically from Meta Business Manager into the platform (and vice versa).

6.5.3 Broadcast & Campaigns

- Send broadcast campaigns to segmented subscriber lists (e.g., past customers of a specific tour, inactive customers, VIP customers) respecting WhatsApp's 24-hour session and template rules.
- Schedule broadcasts, view delivery/read/response analytics, and manage opt-out/unsubscribe automatically per WhatsApp/Meta policy.

6.5.4 Subscriber & Contact Management

- Central WhatsApp subscriber database with tags, custom fields, consent/opt-in status, and conversation history.
- Import/export/sync contacts between website customer accounts, email subscribers, and WhatsApp contacts (unified customer profile).

6.5.5 Staff Chat Management

- **Shared team inbox:** All incoming WhatsApp conversations appear in one inbox visible to authorized staff
- **Chat assignment:** Conversations can be manually or automatically (round-robin/load-based) assigned to a specific staff member; reassignment supported
- **Labels/tags:** Custom labels (e.g., "New Lead", "Payment Pending", "VIP", "Complaint") for filtering and reporting
- **Internal notes:** Private notes on a conversation visible only to staff, not the customer
- **Canned responses / quick replies:** Predefined answers for FAQs to speed up agent response time
- **Agent performance view:** Response time, resolved chats, and conversion (chats → bookings) per staff member

6.5.6 Automation, Bot Builder & AI Assistant

- **Visual bot/flow builder:** Drag-and-drop conversation flow builder (trigger → condition → action) for FAQs, booking flows, and follow-ups, requiring no coding
- **Keyword & intent triggers:** Bot responds to keywords, button clicks, or detected intent (e.g., "availability", "price", "cancel")
- **AI Assistant (LLM-powered):** Handles natural-language questions ("Do you have a desert safari this weekend for 4 people?"), pulls live data (availability/pricing) via API/function-calling, and can complete a booking conversationally
- **Human handoff:** Bot detects when it cannot resolve a query (or customer requests a human) and hands the chat to an available agent, with full context passed along
- **Business hours & auto-reply:** Away messages outside working hours, with option to still allow AI-handled bookings 24/7

6.6 Email Management

- **Template builder:** Drag-and-drop / HTML email editor for transactional and marketing templates, with variable placeholders and preview
- **Template sync:** Reusable template blocks shared between transactional emails (confirmations, receipts) and marketing campaigns

- **Broadcast/campaign manager:** Segment-based bulk sending, scheduling, A/B subject-line testing, open/click tracking, bounce/complaint handling
- **Subscriber manager:** List management, custom fields, tags, double opt-in, unsubscribe handling, GDPR/PDPL-compliant consent log
- **Transactional email engine:** Order confirmation, payment receipt, booking reminder, cancellation/refund notice — all fully detailed and branded
- **Deliverability:** SPF/DKIM/DMARC configuration, dedicated sending domain, and integration with a transactional ESP (e.g., SendGrid/Postmark/Amazon SES) for reliability

6.7 Unified Order Confirmation Flow

As specifically requested — once a customer submits booking + payment details (online or bank-transfer screenshot) and the admin approves it, the system automatically triggers:

- An Email confirmation with: Order ID, tour name, date/time/slot, number of pax, add-ons, total amount paid, payment method, meeting point/instructions, cancellation policy, support contact, and a downloadable PDF voucher/invoice.
- A WhatsApp confirmation message (using an approved template) with the same key details, optionally including a QR code/voucher image and quick links (Get Directions / Contact Support / Reschedule).
- An internal notification to the assigned staff member / operations team so the field logistics (guide, vehicle) can be arranged.

6.8 Admin / Back-Office Control Panel

A single, full-control dashboard covering:

- Dashboard: real-time KPIs — today's bookings, revenue, pending verifications, open chats, upcoming tours.
- Tour/Service Management: create/edit tours, pricing tiers, media gallery, itinerary, SEO fields, add-ons.
- Slot & Calendar Management (see 6.2).
- Order & Booking Management (see 6.3) with search/filter by status, date, customer, channel.
- Payment Management: gateway settings (AmwalPay keys/webhooks), bank account setup, manual-payment verification queue, refunds, reconciliation reports.
- WhatsApp Manager: template builder, broadcast/campaign manager, subscriber manager, bot/automation builder, shared inbox, labels, staff assignment rules.
- Email Manager: template builder, campaigns, subscriber lists, deliverability settings.
- Staff & Role Management: add staff, assign roles/permissions, set working hours, view performance.
- CRM: unified customer profiles (bookings + WhatsApp + email history in one view), segments/tags.
- Coupons & Promotions: discount codes, seasonal offers, referral codes.
- Reports & Analytics: sales/revenue, top tours, channel performance (Web vs WhatsApp), conversion funnel, campaign performance, staff performance.
- System Settings: languages, currencies, tax/VAT rules, cancellation policy text, notification templates, API keys/integrations, audit logs, backups.
- Content Management: banners, blog/SEO pages, FAQs shown on the website and reusable as WhatsApp quick answers.

7. Suggested Additional Features (Recommended Enhancements)

Based on the requirements shared, here are gaps and value-adding features worth considering before development begins:

- **Digital voucher with QR check-in:** Generate a scannable QR code per booking that field staff can scan on the day of the tour to confirm attendance/no-show — closes the loop between booking and delivery
- **Deposit / partial payment support:** Allow a customer to pay a percentage upfront and balance later, common for tour bookings — reduces drop-off at checkout
- **Dynamic/seasonal pricing rules:** Auto-adjust prices by date, demand, or remaining capacity (e.g., higher price on weekends/holidays)
- **Waitlist & auto-notify:** When a slot is full, let customers join a waitlist and get an automatic WhatsApp/email alert if a seat opens up (cancellation)
- **Reviews & ratings with photo upload:** Post-tour review requests (auto-sent via WhatsApp/email) feed public star ratings on the tour page — builds trust and SEO
- **Referral & loyalty program:** Reward repeat customers or referrals with discount codes distributed via WhatsApp/email
- **Abandoned booking recovery:** Auto-detect incomplete bookings (slot selected, payment not completed) and send a reminder via WhatsApp/email with a direct resume link
- **Fraud/duplicate-screenshot detection:** Basic checks (hash comparison, OCR of amount/reference vs submitted order) to flag suspicious manual-payment submissions before admin review
- **Multi-branch / multi-location support:** If the business operates in more than one city/branch, structure data so each branch has its own inventory, staff, and bank account, with combined reporting

- **Guide/vehicle resource management:** Beyond "slots", track which guide/vehicle is assigned to each departure to avoid double-booking staff/resources
- **Customer self-service reschedule/cancel:** Allow customers to reschedule or cancel within policy directly from WhatsApp or the website, subject to rules the admin sets
- **Multi-agent / call-center style SLA alerts:** Auto-escalate a WhatsApp chat if unanswered beyond X minutes (customer-experience protection)
- **Data export & accounting integration:** Export financial data to formats compatible with common accounting tools; consider VAT-compliant invoicing for Oman (5% VAT) from day one
- **Native app or PWA (Phase 2):** A Progressive Web App wrapper gives an app-like experience (push notifications, home-screen install) with less cost than native apps
- **Disaster recovery & backups:** Automated daily database backups with a tested restore procedure — critical since this handles payments and customer data
- **Consent & data privacy compliance:** Explicit opt-in capture for WhatsApp/email marketing and a privacy policy aligned with Oman's Personal Data Protection Law (PDPL)

8. Non-Functional Requirements

Category	Requirement
Performance	Website pages load in <3 seconds on 4G; availability check API responds in <1 second; system supports at least 500 concurrent users at launch, horizontally scalable beyond that.
Availability	Target 99.9% uptime; payment and WhatsApp webhook endpoints must have retry/queue handling to avoid lost events.
Security	HTTPS everywhere, PCI-DSS-aware handling of payment data (tokenized where possible, no raw card storage), encrypted storage of screenshots/personal data, role-based access control, audit logging of admin actions.
Scalability	Cloud-hosted, containerized architecture that can scale booking, WhatsApp, and email services independently as volume grows.
Localization	English and Arabic (RTL support) at minimum; OMR currency by default with multi-currency ready architecture.
Compliance	Oman PDPL data-protection alignment; WhatsApp Business Policy and Meta Commerce Policy compliance for template/marketing messages; VAT-compliant invoicing.
Maintainability	Modular codebase (booking, payments, WhatsApp, email, admin as separate services/modules) with API documentation for future integrations.
Auditability	Every payment approval/rejection, booking modification, and refund is logged with user, timestamp, and reason.

9. Recommended Technical Architecture

9.1 Suggested High-Level Stack

Layer	Recommendation
Frontend (Website)	Next.js / React (SEO-friendly, fast, PWA-capable) with a headless component library; Tailwind CSS for styling.
Admin Panel	React-based SPA (or same Next.js app under /admin) with role-based routing.
Backend / API	Node.js (NestJS) or Laravel/PHP — REST + webhook endpoints, with clear service boundaries: Booking Service, Payment Service, WhatsApp Service, Email Service, Notification Service.
Database	PostgreSQL (relational, strong consistency for bookings/payments) + Redis (caching, slot-locking during checkout, session queues).
File/Media Storage	Cloud object storage (e.g., S3-compatible) for tour images, payment screenshots, and generated vouchers/invoices.
WhatsApp Integration	Official WhatsApp Cloud API via a licensed BSP (e.g., a Meta Business Partner) — do NOT use unofficial/automated WhatsApp libraries, which risk number bans.
Payment Gateway	AmwalPay hosted checkout/API with signed webhook verification; PCI-DSS SAQ-A scope by using hosted/redirect checkout rather than storing card data.

Layer	Recommendation
Email Delivery	Transactional ESP (e.g., Amazon SES, SendGrid, or Postmark) integrated with the in-house template/broadcast manager.
AI Assistant	LLM API (e.g., Claude via Anthropic API) with function-calling into the Booking/Order APIs for live availability and booking actions, plus guardrails for handoff to humans.
Hosting/Infra	Cloud provider (AWS/Azure/GCP) with containerized services (Docker + orchestration), CDN for static/media assets, automated backups and monitoring/alerting (e.g., Grafana, Sentry).
Queue/Jobs	Message queue (e.g., RabbitMQ/SQS) for WhatsApp/email sending, webhook processing, and reminders — ensures nothing is lost under load.

9.2 Key Integration Flows

- **AmwalPay flow:** Checkout → create payment session via AmwalPay API → redirect/host page → customer pays → AmwalPay sends signed webhook → system verifies signature → order status updated → confirmation triggered
- **Manual bank transfer flow:** Checkout → customer selects bank transfer → sees account details → submits reference + screenshot → order created as Pending Verification → admin dashboard alert → admin approves/rejects → confirmation or rejection notice triggered
- **WhatsApp booking flow:** Customer message → Cloud API webhook → bot/flow engine (or AI assistant) → calls internal Booking API for availability/creation → payment link or bank-transfer instructions sent in chat → screenshot received as media message → attached to order automatically

10. Core Data Model (Key Entities)

Entity	Key Fields (illustrative)
Tour/Service	id, name, description, category, city, base price, pricing tiers, media, itinerary, status
Slot	id, tour_id, date, time, capacity, seats_booked, resource_id (guide/vehicle), status
Order/Booking	id, customer_id, tour_id, slot_id, pax, add_ons, total_amount, payment_method, payment_status, order_status, channel (web/whatsapp/admin), created_at
Payment	id, order_id, method (amwalpay/bank_transfer), amount, gateway_reference, screenshot_url, verified_by, verified_at, status
Customer	id, name, phone, email, whatsapp_opt_in, email_opt_in, tags, booking_history
Staff/User	id, name, role, permissions, assigned_chats, working_hours
WhatsApp Conversation	id, customer_id, assigned_staff_id, labels, last_message_at, status (open/closed), bot_active (y/n)
Template (WhatsApp/Email)	id, type, category, content, media, variables, approval_status, version
Campaign/Broadcast	id, channel, template_id, segment_id, scheduled_at, stats (sent/delivered/read/clicked)
Subscriber List/Segment	id, name, channel, filter_rules, contact_count

11. Role & Permission Matrix (Illustrative)

Module	Super Admin	Ops/Booking Admin	Finance Verifier	Chat Agent	Marketing
Tours/Slots	Full	Full	View	View	View
Orders	Full	Full	View/Update Payment	View	View
Payment Verification	Full	View	Full	—	—
WhatsApp Inbox	Full	View	—	Assigned Chats	Broadcast Only
Templates (Email/WA)	Full	—	—	—	Full
Staff Management	Full	—	—	—	—
Reports	Full	Bookings Only	Payments Only	Own Performance	Campaigns Only
System Settings	Full	—	—	—	—

12. Assumptions & Constraints

- This is treated as a single-business (not multi-vendor) platform unless clarified otherwise.
- AmwalPay merchant account and API credentials will be provided by the business/AmwalPay onboarding team.
- WhatsApp Cloud API access requires a verified Meta Business Manager account, a dedicated business phone number, and either direct Cloud API access or a licensed BSP — this has an approval lead time and template-approval process outside the development team's control.
- AI Assistant responses must stay within business-provided data (tours, pricing, policies) to avoid giving incorrect commitments to customers.
- Company bank account details for manual transfer will be provided by finance and are editable only by Super Admin.
- Primary languages assumed as English and Arabic; other languages can be added later.

13. Suggested Project Phases & Milestones

Phase	Deliverables	Indicative Focus
Phase 0 – Discovery & UX	Finalized BRD sign-off, wireframes, information architecture, data model	Requirements lock, design system
Phase 1 – Core Booking + Manual Payment	Website booking flow, slot engine, order management, bank-transfer + screenshot verification, admin panel v1, email confirmations	Launch-ready MVP
Phase 2 – WhatsApp Commerce	WhatsApp Cloud API integration, availability/booking/order-status in chat, template manager, shared inbox, staff assignment/labels	Conversational commerce
Phase 3 – Automation & AI	Bot/flow builder, AI assistant with function-calling, broadcast/campaign manager (WhatsApp + Email), subscriber segmentation	Automation & marketing
Phase 4 – AmwalPay & Growth Features	AmwalPay online payment integration, coupons, waitlist, reviews, reporting/analytics dashboard, PWA	Growth & optimization

14. Success Metrics / KPIs

- Online booking conversion rate (visits → completed bookings).
- % of bookings completed via WhatsApp vs Website.
- Average payment verification turnaround time (manual bank transfer).
- WhatsApp bot resolution rate (queries resolved without human handoff).
- Average agent response time and chats-to-booking conversion.
- Email/WhatsApp campaign open, click, and conversion rates.
- Repeat booking rate / customer retention.

15. Risks & Mitigation

Risk	Mitigation
WhatsApp template rejected by Meta or number gets flagged for spam-like broadcasting	Follow Meta's messaging policy strictly, keep opt-in records, avoid promotional content in utility templates, monitor quality rating
Fraudulent/edited payment screenshots submitted	Require transaction reference + screenshot, cross-check against bank statement during reconciliation, restrict who can approve
Double-booking due to concurrent checkout on same slot	Slot-locking mechanism (temporary hold with Redis/DB lock) during checkout window
AI assistant gives inaccurate info or overcommits pricing/availability	Restrict AI to real-time function calls into the booking API rather than free-form answers on price/availability; human-handoff for edge cases
Payment gateway or WhatsApp API downtime	Queue-based retry mechanism, fallback manual payment option always available, status page/alerting for admin
Delay in AmwalPay/WhatsApp Business approval timelines	Start account/API onboarding in parallel with Phase 0/1 development, not after

16. Appendix — Glossary

Term	Meaning
BSP	Business Solution Provider — a Meta-authorized partner that provides WhatsApp Cloud API access and tooling
Cloud API	Meta's official WhatsApp Business API used for automated, programmatic messaging
AmwalPay	Payment gateway/PSP used for online card and local payment processing
Slot	A bookable date/time unit of a tour with limited capacity
PDPL	Oman's Personal Data Protection Law governing customer data handling
BRD	Business Requirements Document — this document